

Differential gene expression analysis in HCC

Tumor vs adjacent non-tumor liver tissues • GSE22058

Aim: identify differentially expressed genes and interpret their biological and miRNA-target regulation patterns.

15-minute presentation

Paired design

Functional enrichment



Study design and biological question

Question

Which genes and biological processes distinguish HCC tumor tissue from adjacent non-tumor liver tissue?

Tumor



Adjacent non-tumor



96

paired patients

192

samples analyzed

2

conditions

Paired comparison

- Tumor and adjacent non-tumor tissue come from the same individuals.
- This controls part of the inter-patient variability.
- Therefore, a paired t-test is appropriate downstream.

Analysis workflow



Import

GEO matrix + metadata



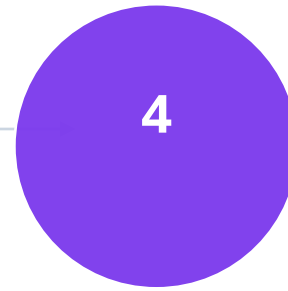
Preprocess

$\log_2(\text{data} + 1)$ + IQR filter



Test

paired t-test + FDR



Visualize

volcano, boxplots, heatmap,
PCA



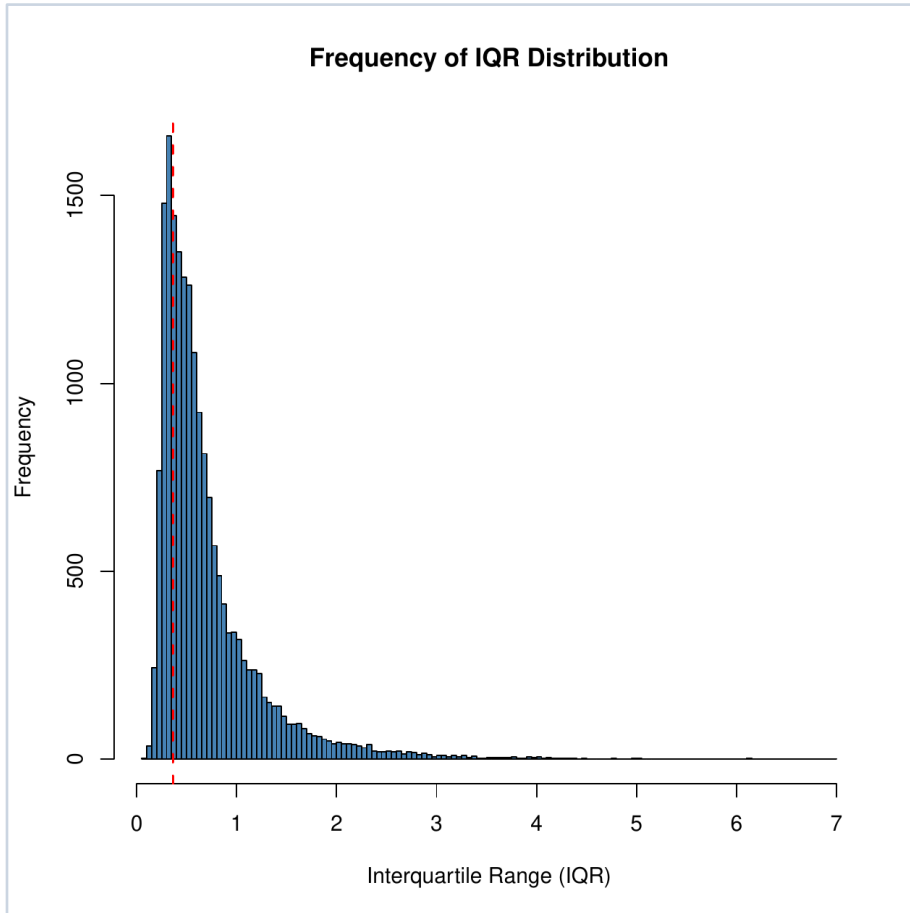
Interpret

EnrichR + MIENTURNET

Output of the pipeline

- 1667 differentially expressed genes
- Directional pathway enrichment
- Candidate miRNA-target regulatory modules

Preprocessing: remove low-information genes



IQR filtering

- Expression values were transformed as $\log_2(\text{data} + 1)$.
- IQR was computed for each gene across all samples.
- Genes below the 25th percentile of IQR were removed.
- This keeps genes with enough variability to be informative.

18,503

genes before filtering

13,877

genes retained

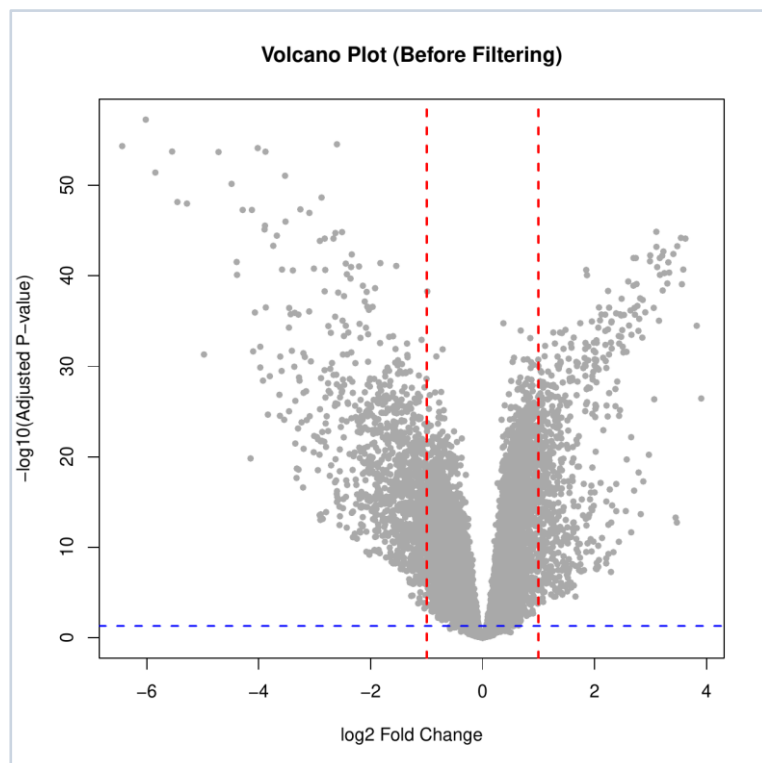
25%

low-IQR genes removed

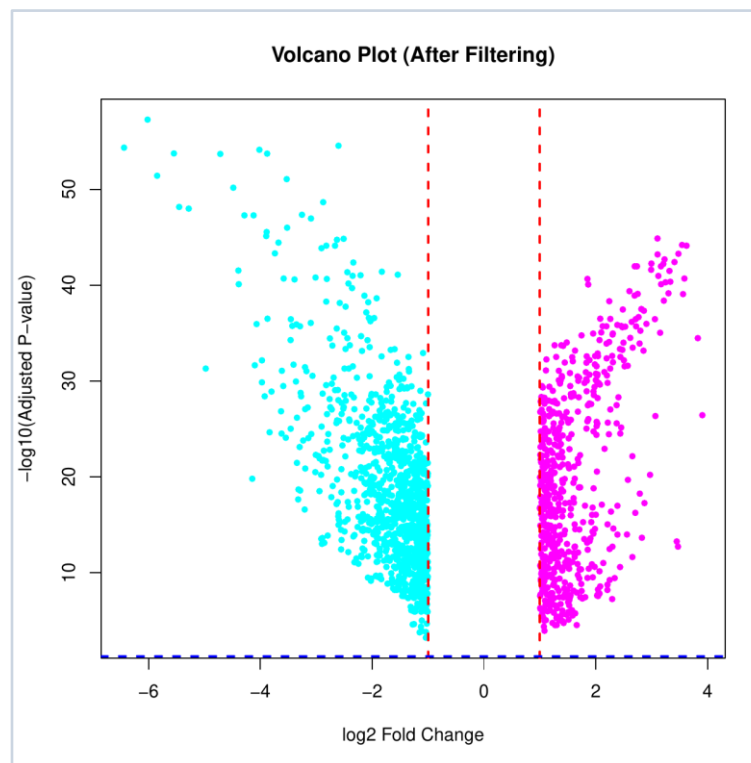
Interpretation

The histogram is highly concentrated near low IQR values, so filtering removes genes with little variation while preserving most of the transcriptomic signal.

Differential expression: 1667 genes pass FC and FDR thresholds



Before filtering



After filtering

Thresholds

- Fold-change ≥ 2
- $|\log_2FC| \geq 1$
- Adjusted p-value < 0.05
- Paired Student's t-test

1667

DEGs

602

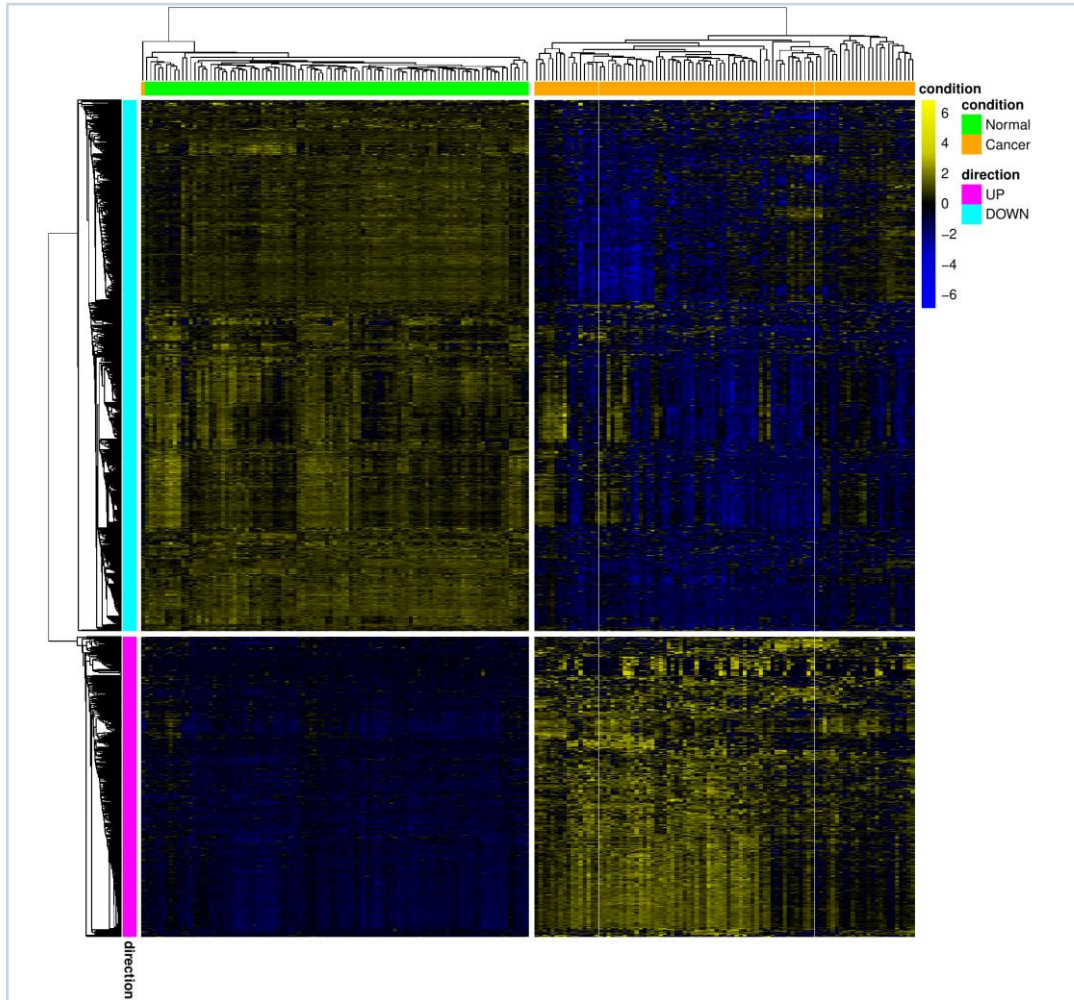
UP-regulated

1065

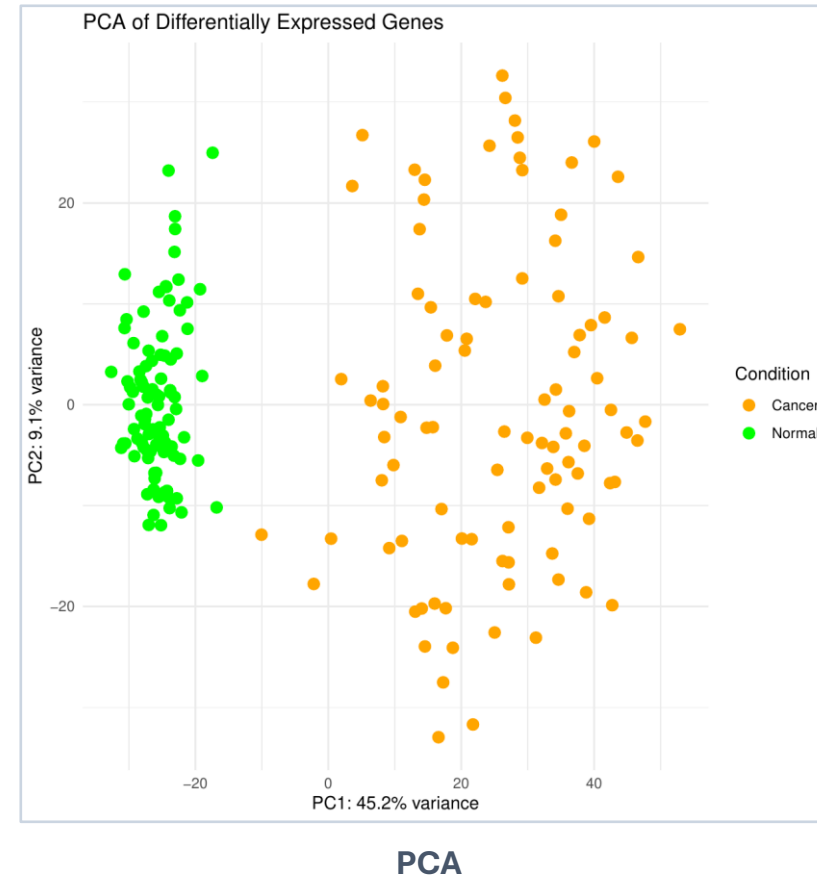
DOWN-regulated

Positive logFC = higher expression in tumor.
Negative logFC = lower expression in tumor.

DEG profiles separate tumor from adjacent normal samples



Heatmap

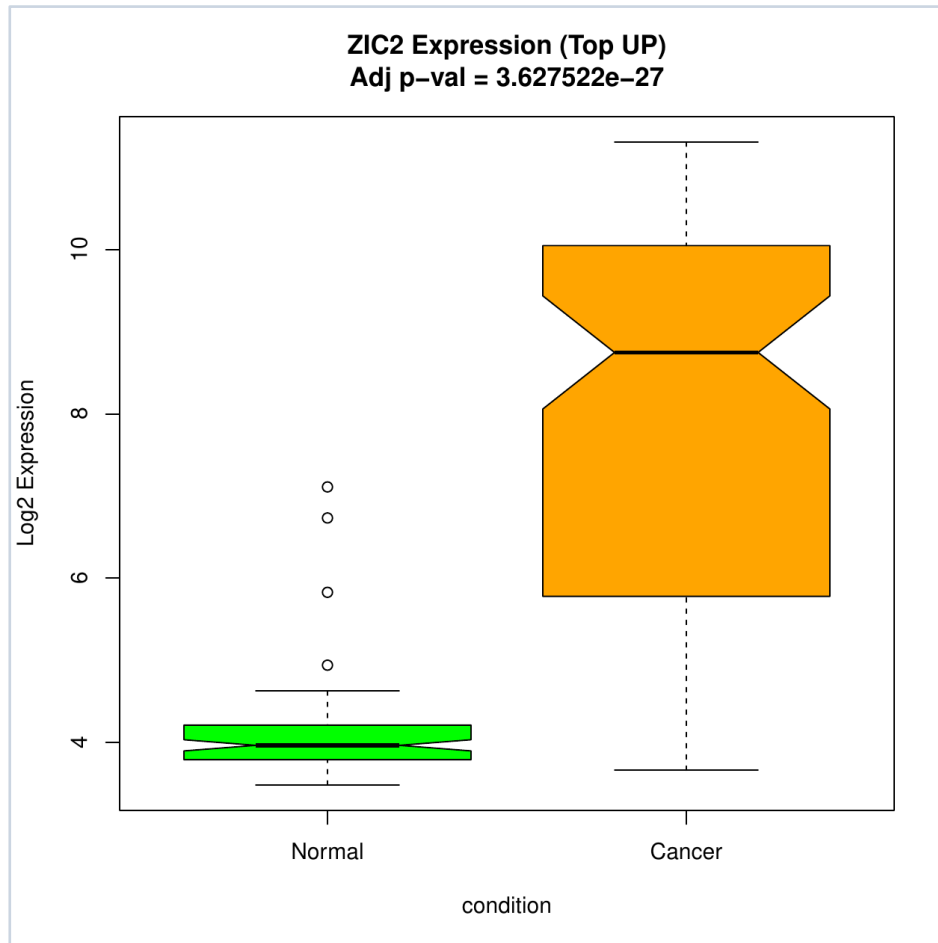


Key readout

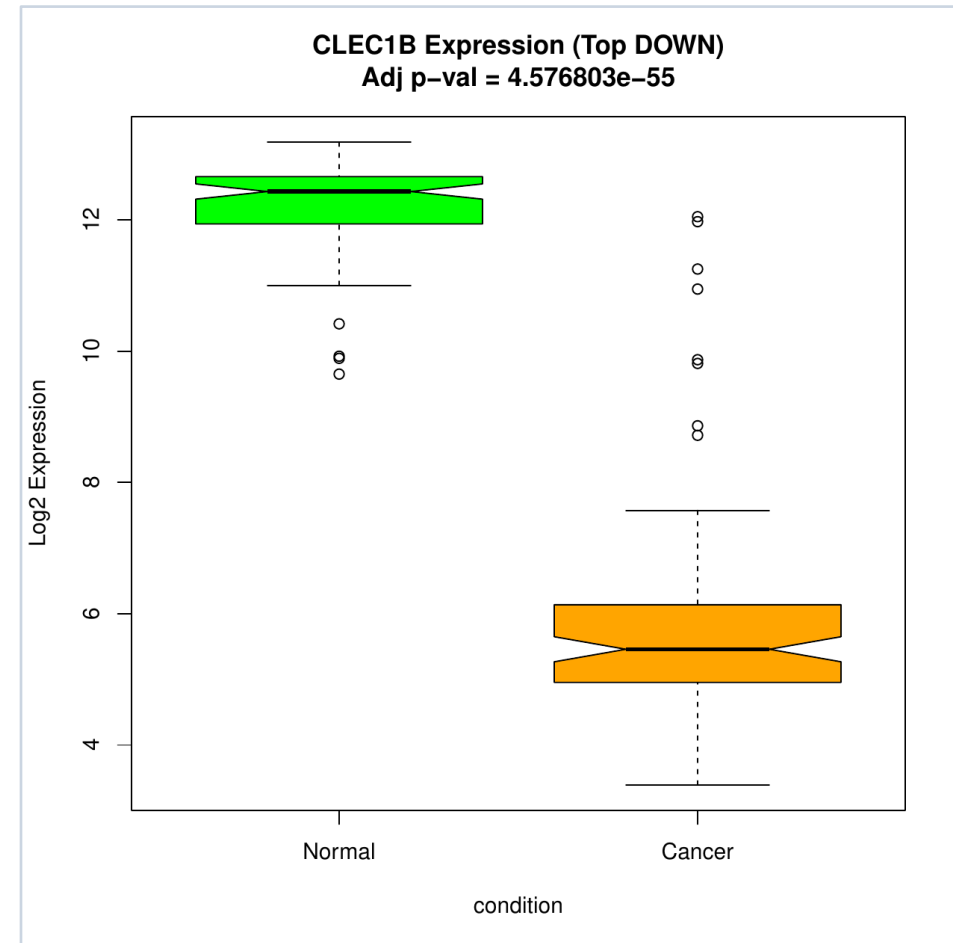
PC1 explains 45.2% of variance and clearly separates normal and tumor samples.

The heatmap shows opposite expression blocks for UP and DOWN genes.

Representative genes validate the direction of change

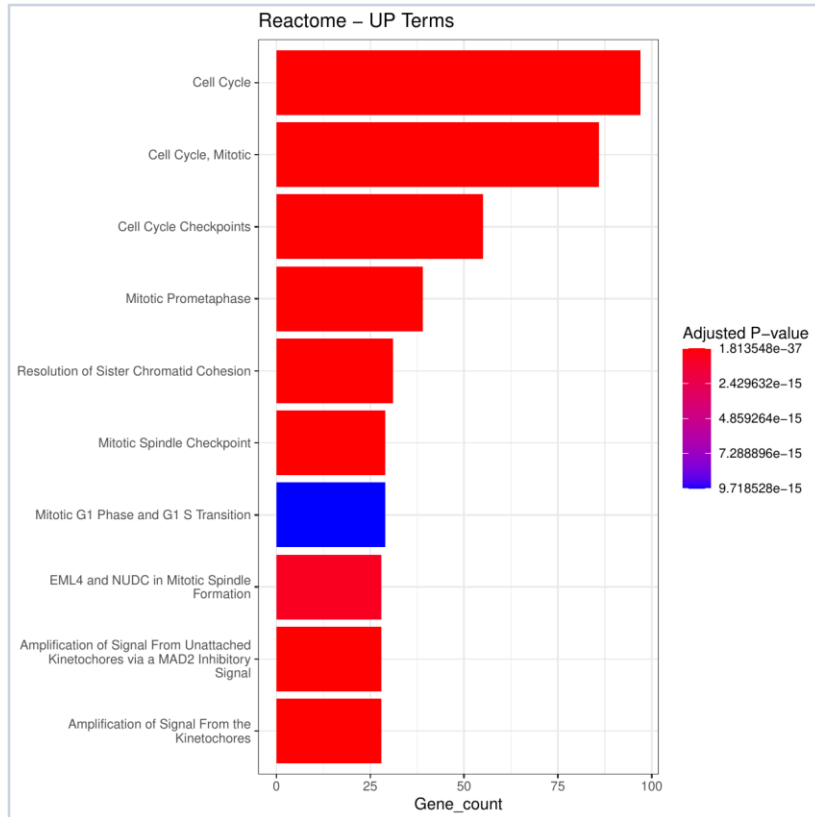


ZIC2: top UP-regulated gene

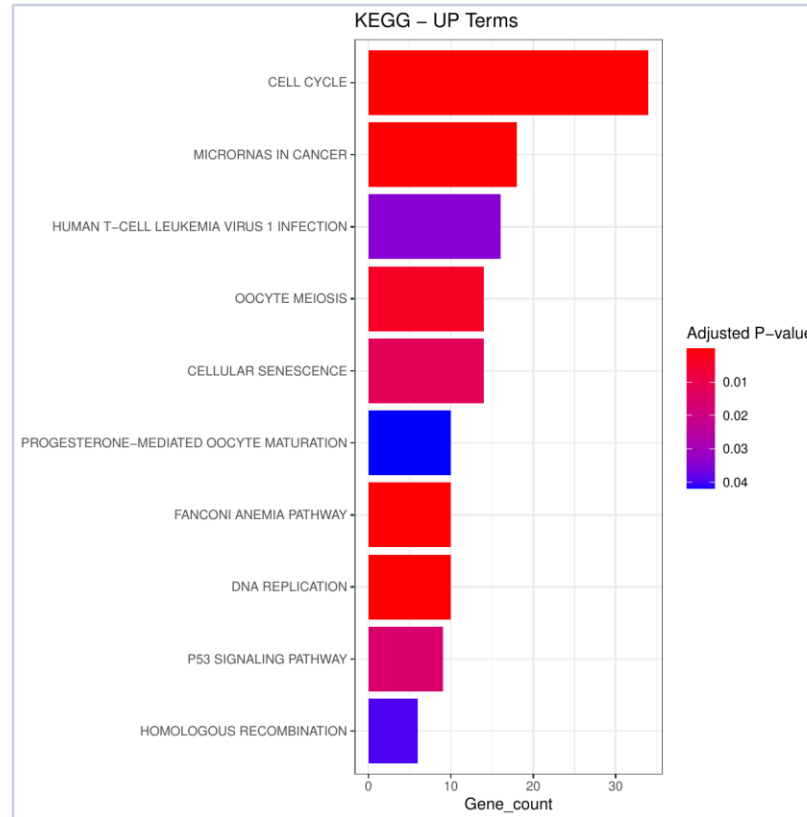


CLEC1B: top DOWN-regulated gene

UP-regulated genes: cell cycle and mitotic proliferation



Reactome



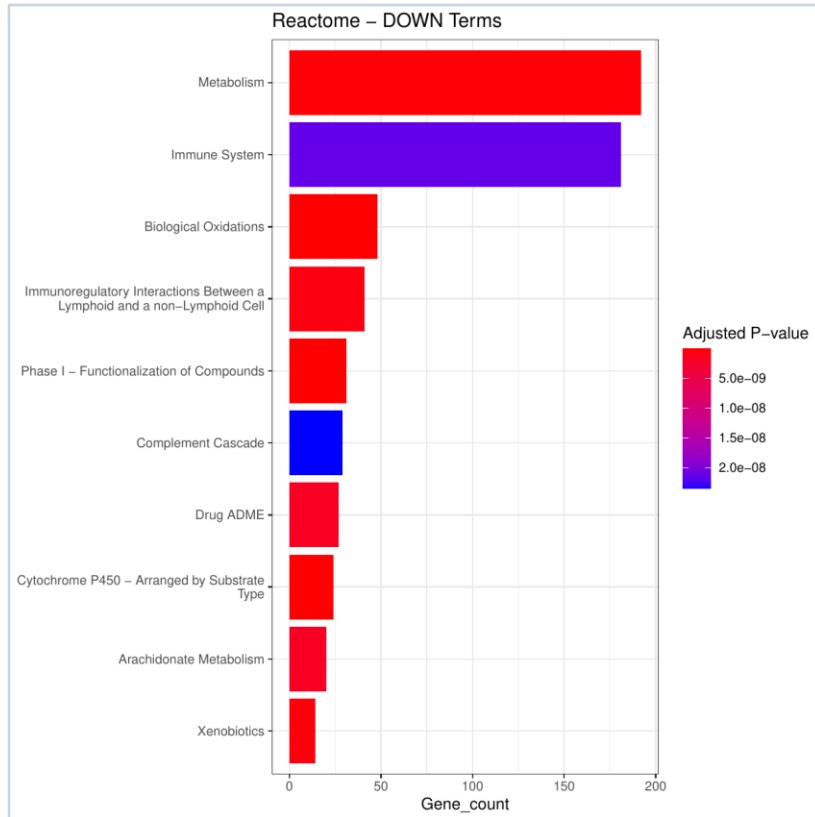
KEGG

Biological message

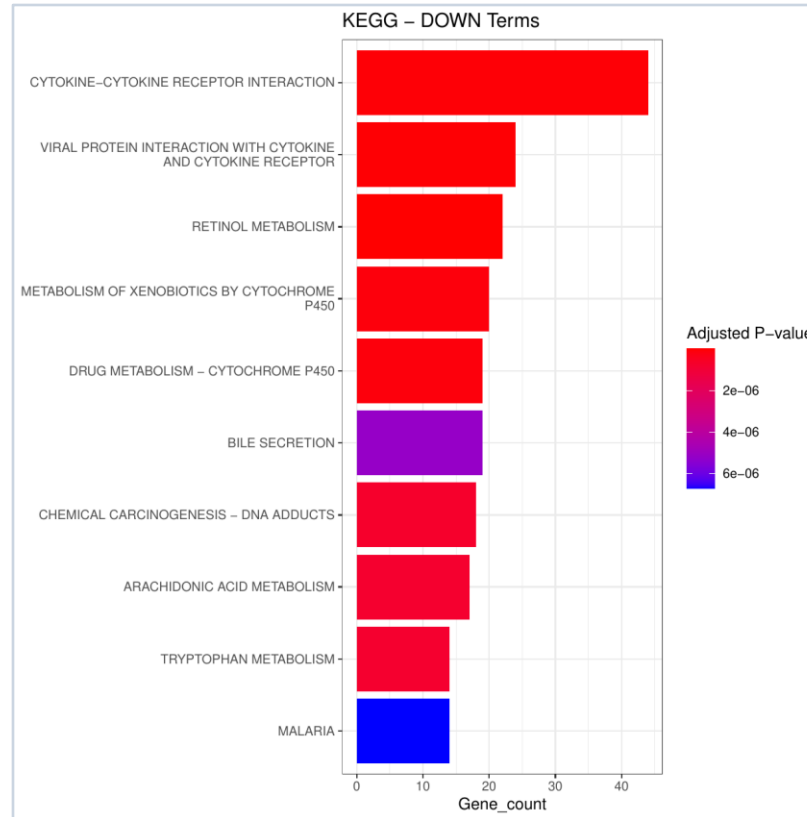
- Cell cycle
- DNA replication
- Mitotic checkpoints
- Chromosome segregation
- p53 signaling
- MicroRNAs in cancer

Tumor samples activate proliferative programs.

DOWN-regulated genes: loss of liver metabolism and immune remodeling



Reactome



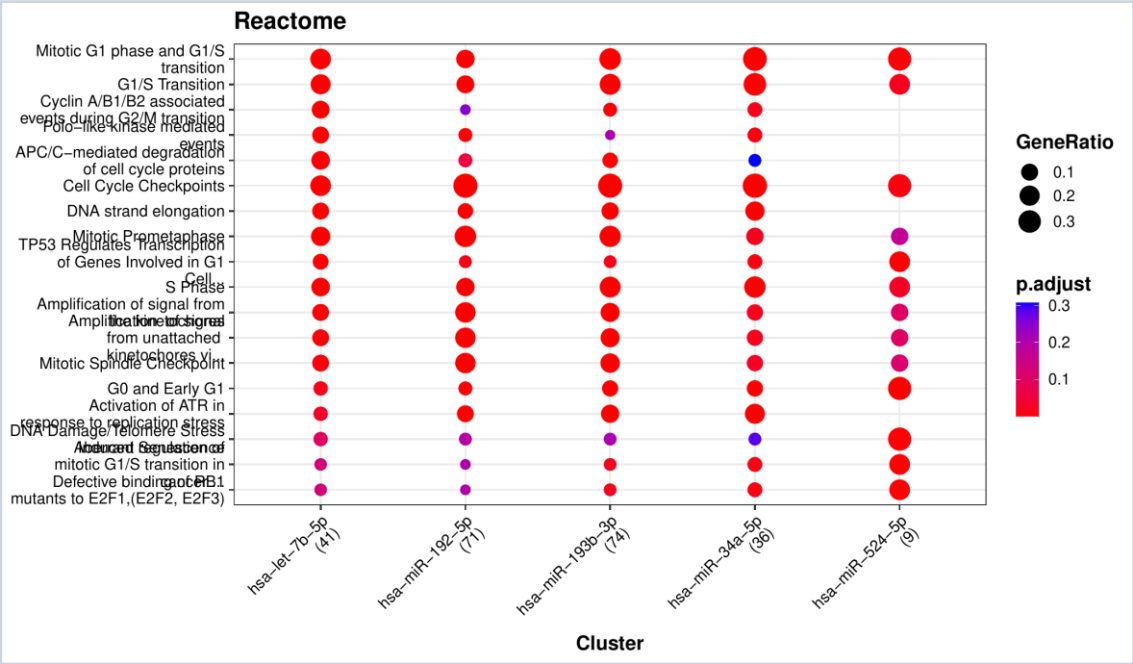
KEGG

Biological message

- Cytochrome P450 metabolism
- Xenobiotic and drug metabolism
- Retinol metabolism
- Bile secretion
- Biological oxidations
- Cytokine/chemokine signaling

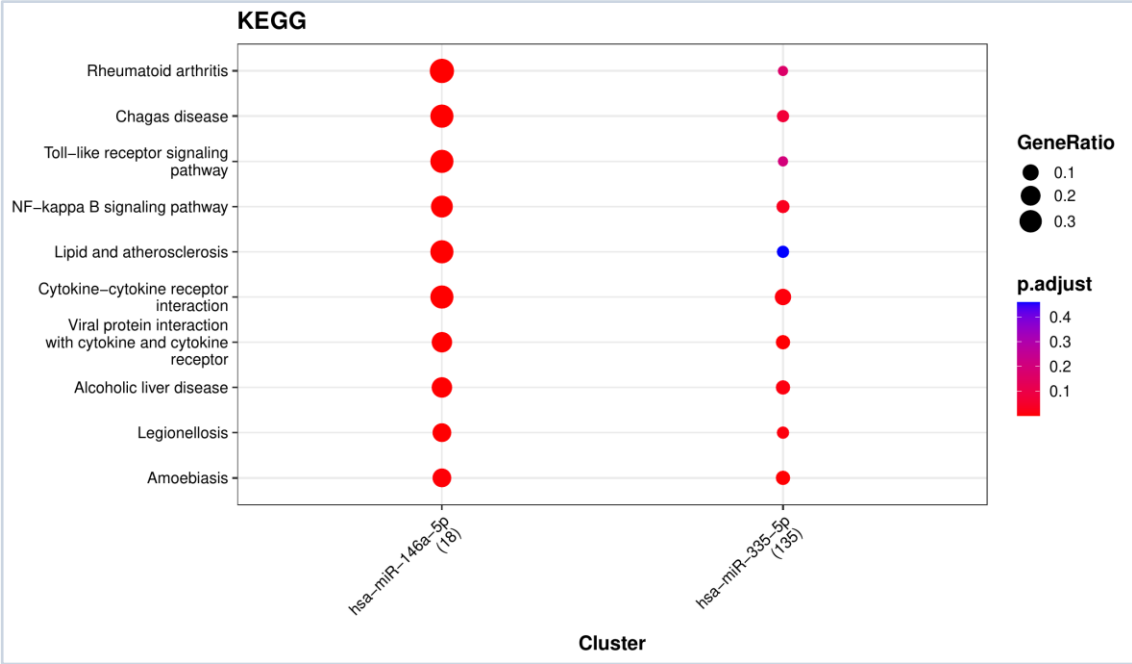
Tumor samples lose normal hepatocyte metabolic functions.

MIENTURNET: candidate miRNA regulators of DEG modules



UP genes: cell-cycle/DNA replication miRNA-target modules

- hsa-let-7b-5p
- hsa-miR-192-5p
- hsa-miR-193b-3p
- hsa-miR-34a-5p



DOWN genes: immune/inflammatory miRNA-target modules

- hsa-miR-146a-5p
- hsa-miR-335-5p

Interpret with caution: these are enriched candidate regulators, not measured miRNA expression changes.

Literature-based validation of selected findings

Purpose: check whether the main computational findings are biologically plausible according to previous HCC/cancer literature.

ZIC2: top UP-regulated gene by logFC

Reported as a prognostic and immune-response marker in liver cancer/HCC.

Supports the relevance of the strong tumor-up signal.

CLEC1B: top DOWN-regulated gene by logFC

Reported as a prognostic biomarker in HCC and associated with immune infiltration.

Consistent with its strong reduction in tumor samples.

CYP450 / drug metabolism loss

HCC literature reports down-regulation/deregulation of xenobiotic-metabolizing CYP genes.

Supports loss of normal hepatocyte detoxification function.

miRNA-target findings

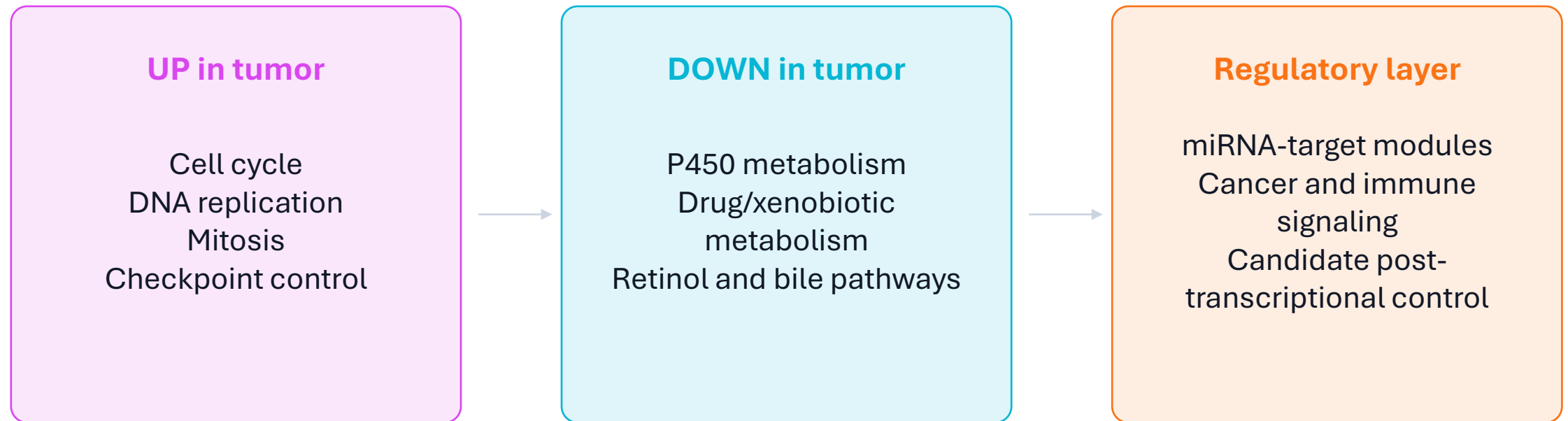
miR-146a: immune/inflammatory and TLR/NF-kB-related signaling.

miR-34a: p53-related miRNA linked to cell-cycle arrest, DNA-damage response, and apoptosis.

Conclusion: findings are consistent with published HCC/cancer biology, but remain hypothesis-generating.

Selected examples: Sun et al. 2021; Liang et al. 2022; Nekvindova et al. 2020; Fu et al. 2023.

Integrated biological interpretation



Overall conclusion

HCC tumor tissue is characterized by activation of proliferative cell-cycle programs and reduced expression of normal liver metabolic/detoxification functions, with additional immune and miRNA-target regulatory signals.

Conclusions and limits

Main conclusions

- 1667 DEGs distinguish tumor from adjacent non-tumor liver samples.
- UP-regulated genes point to proliferation, DNA replication, and mitotic checkpoint activity.
- DOWN-regulated genes point to loss of hepatocyte metabolism and detoxification.
- MIENTURNET suggests candidate miRNA regulators of proliferative and immune/inflammatory modules.

Limitations

- miRNAs are inferred from target enrichment, not directly tested for expression.
- Functional enrichment is hypothesis-generating.
- Further validation would require independent cohorts or experimental follow-up.

Take-home message

The pipeline produced a biologically coherent HCC signature: proliferative activation in tumors, loss of normal liver function, and plausible miRNA-target regulatory modules.